

CBSE Class 12 Chemistry — Chapterwise Formula Sheet

As per the latest CBSE 2025-26 syllabus, chapters like Solid State, Surface Chemistry, and Polymers have been completely deleted [1, 2]. The formula sheet below focuses strictly on the 10 units prescribed in the current curriculum, emphasizing the heavily tested numerical chapters (Physical Chemistry) and key relations in Inorganic and Organic Chemistry [3, 4].

1. Solutions

Concentration Terms * **Molarity (M):** $M = \frac{\text{Moles of Solute}}{\text{Volume of Solution (in L)}} [5]$ * **Molality (m):** $m = \frac{\text{Moles of Solute}}{\text{Mass of Solvent (in kg)}} [5]$ * **Mole Fraction (χ):** $\chi_A = \frac{n_A}{n_A + n_B} [5]$

Solubility and Vapour Pressure * **Henry's Law:** $P_{gas} = K_H \cdot \chi_{gas}$ (where P_{gas} is partial pressure, K_H is Henry's constant) [6]. * **Raoult's Law (Volatile Solute):**

$$P_{Total} = P_A + P_B = (P_A^\circ \cdot \chi_A) + (P_B^\circ \cdot \chi_B) [7, 8].$$

Colligative Properties & Van't Hoff Factor (i) * **Relative Lowering of Vapour Pressure**

(RLVP): $\frac{P_{solvent}^\circ - P_{solution}}{P_{solvent}^\circ} = i \cdot \chi_{solute} [8]$. * **Elevation in Boiling Point (ΔT_b):** $\Delta T_b = i \cdot K_b \cdot m$

[5, 8]. * **Depression in Freezing Point (ΔT_f):** $\Delta T_f = i \cdot K_f \cdot m [5, 8]$. * **Osmotic Pressure (Π):** $\Pi = i \cdot C \cdot R \cdot T [5, 8]$.

Degree of Association/Dissociation * **Degree of Dissociation (α):** $\alpha = \frac{i-1}{n-1}$ (where n is the number of ions produced per formula unit) [9]. * **Degree of Association (α):** $\alpha = \frac{1-i}{1-1/n}$ (where n is the number of molecules associating).

Important Constants: * Universal Gas Constant (R): $0.0821 \text{ L atm K}^{-1} \text{ mol}^{-1}$ or $8.314 \text{ J K}^{-1} \text{ mol}^{-1} [10, 11]$.

2. Electrochemistry

Conductance and Resistance * **Conductance (G):** $G = \frac{1}{R} [5]$. * **Cell Constant (G^*):** $G^* = \frac{l}{A} [4, 12]$. * **Specific Conductivity (κ):** $\kappa = G \cdot G^* = \frac{1}{R} \cdot \frac{l}{A} [4, 5]$. * **Molar Conductivity (Λ_m):** $\Lambda_m = \frac{\kappa \times 1000}{C}$ (where C is Molarity, unit is $\text{S cm}^2 \text{ mol}^{-1}$) [5].

Kohlrausch's Law & Weak Electrolytes * **Limiting Molar Conductivity:**

$\Lambda_m^\circ(AB) = \lambda_+^\circ + \lambda_-^\circ [13]$. * **Degree of Dissociation (α):** $\alpha = \frac{\Lambda_m}{\Lambda_m^\circ} [14]$. * **Acid Dissociation Constant (K_a):** $K_a = \frac{C\alpha^2}{1-\alpha} [5, 14]$.

Cell Potential and Nernst Equation * **Standard Cell Potential:** $E_{cell}^\circ = E_{cathode}^\circ - E_{anode}^\circ [5, 13]$. * **Nernst Equation (at 298 K):** $E_{cell} = E_{cell}^\circ - \frac{0.0591}{n} \log Q [5]$. * **Nernst Equation (General):** $E_{cell} = E_{cell}^\circ - \frac{2.303RT}{nF} \log \frac{[\text{Products}]}{[\text{Reactants}]} [9, 15]$.

Thermodynamics of Cell * Gibbs Free Energy: $\Delta G^\circ = -nFE_{cell}^\circ$ [5]. * **Equilibrium Constant (K_c):** $E_{cell}^\circ = \frac{0.0591}{n} \log K_c$ or $\Delta G^\circ = -2.303RT \log K_c$ [16, 17].

Faraday's Laws of Electrolysis * First Law: $W = Z \cdot Q = \frac{M}{nF} It$ (where W is mass deposited, I is current in amperes, t is time in seconds) [13, 18]. * **Second Law:** $\frac{W_1}{W_2} = \frac{E_1}{E_2}$ (where E is equivalent weight) [18].

Important Constants: * Faraday Constant (F): 96500 C mol^{-1} [16].

3. Chemical Kinetics

Rate of Reaction * Rate Law: $\text{Rate} = k[A]^m[B]^n$ [17, 19]. * **Overall Order:** $x = m + n$ [20].

Integrated Rate Equations & Half-Life * Zero Order Reaction: * Integrated rate: $k = \frac{[A]_0 - [A]}{t}$ [19, 20]. * Half-life ($t_{1/2}$): $t_{1/2} = \frac{[A]_0}{2k}$ [19]. * **First Order Reaction:** * Integrated rate: $k = \frac{2.303}{t} \log \frac{[A]_0}{[A]}$ [19]. * Half-life ($t_{1/2}$): $t_{1/2} = \frac{0.693}{k}$ [17, 19].

Arrhenius Equation (Effect of Temperature) * Exponential Form: $k = Ae^{-E_a/RT}$ [17, 19]. * **Logarithmic Form:** $\log k = \log A - \frac{E_a}{2.303RT}$ [19]. * **Comparing two temperatures:** $\log \left(\frac{k_2}{k_1} \right) = \frac{E_a}{2.303R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$ [21].

4. d- and f-Block Elements

- **Spin-Only Magnetic Moment (μ):** $\mu = \sqrt{n(n+2)} \text{ BM}$ (where n is the number of unpaired electrons) [22].
- **General Electronic Configuration:**
 - Transition Metals: $(n-1)d^{1-10}ns^{1-2}$ [23, 24].
 - Lanthanoids: $4f^{1-14}5d^{0-1}6s^2$ [23].
- **Important Balanced Redox Equations (Acidic Medium):**
 - Dichromate: $Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$ [25].
 - Permanganate: $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$ [26].

5. Coordination Compounds

- **Crystal Field Splitting Energy Relation:** $\Delta_t = \frac{4}{9} \Delta_o$ (Tetrahedral splitting is roughly 4/9ths of octahedral splitting) [27].
- **Crystal Field Stabilization Energy (CFSE) in Octahedral Field:** $CFSE = (-0.4 \cdot t_{2g} + 0.6 \cdot e_g) \Delta_o$ [27, 28].

- **Magnetic Behavior Rules (VBT):**

- Strong field ligands (e.g., CO , CN^- , en , NH_3) typically pair up electrons forming low-spin, inner-orbital complexes [29, 30].
- Weak field ligands (e.g., F^- , Cl^- , H_2O) avoid pairing, forming high-spin, outer-orbital complexes [30].

6. Haloalkanes and Haloarenes to 10. Biomolecules (Organic Chemistry)

Organic chemistry in Class 12 heavily relies on mechanisms and named reactions rather than mathematical formulas, but the following are the most critical quantifiable and structural rules to memorize:

Haloalkanes & Haloarenes * **Chirality & Optical Rotation:** Specific rotation $[\alpha] = \frac{\alpha_{obs}}{l \cdot c}$ [31].

* **S_N1 vs S_N2 Kinetics:** * **S_N1 Rate:** $\text{Rate} = k[\text{Alkyl Halide}]$ (Unimolecular, depends on Carbocation stability: $3^\circ > 2^\circ > 1^\circ$) [32]. * **S_N2 Rate:**

$\text{Rate} = k[\text{Alkyl Halide}][\text{Nucleophile}]$ (Bimolecular, depends on Steric Hindrance: $1^\circ > 2^\circ > 3^\circ$) [33].

Alcohols, Phenols, and Ethers * **Acidity Constant (K_a):** $pK_a = -\log K_a$. Lower pK_a means stronger acid (Phenols are more acidic than alcohols due to resonance stabilization of the phenoxide ion) [34, 35].

Aldehydes, Ketones, and Carboxylic Acids * **Acidity Trends:** Electron-withdrawing groups ($-I$, $-R$) increase acidity (lower pK_a), while electron-donating groups ($+I$, $+R$) decrease acidity [36, 37].

Amines * **Basicity Constant (K_b):** $pK_b = -\log K_b$. Lower pK_b means stronger base. * **Aqueous Phase Basicity Order (Methylamines):**

$(CH_3)_2NH > CH_3NH_2 > (CH_3)_3N > NH_3$ [38]. * **Aqueous Phase Basicity Order (Ethylamines):** $(C_2H_5)_2NH > (C_2H_5)_3N > C_2H_5NH_2 > NH_3$ [39].

Biomolecules * **Carbohydrates:** Number of stereoisomers = 2^n (where n is the number of chiral centers). * **Zwitterion:** Amino acids form dipolar ions where the net charge is zero at the isoelectric point (pI) [40, 41].

CBSE Class 12 (2027) — AI-Predicted via PYQ/Blueprint Analysis. Probability-weighted guidance, not actual exam questions.